Conditions for @Async Annotation to work properly?

1. Different Class :
   If @Aync annotation is applied to the method within the same class from which it is being called, then Proxy mechanism is skipped because internal method calls are **NOT INTERCEPTED**.

2. Public method:
   Method annotated with @Async must be public. And again, AOP interception works only on Public methods.

Both in same class, Proxy will get bypassed

```java
@RestController
@RequestMapping(value = "/api/")
public class UserController {

    @GetMapping(path = "/getuser")
    public String getUserMethod(){
        System.out.println("inside getUserMethod: " + Thread.currentThread().getName() );
        asyncMethodTest();
        return null;
    }

    @Async
    public void asyncMethodTest() {
        System.out.println("inside asyncMethodTest: " + Thread.currentThread().getName());
    }
}
```

This is wrong way of doing it, correct way is, @Async method should be in different class and should be public

Output:

```
2024-08-18T12:46:04.532+05:30  INFO 60618 ---
inside getUserMethod: http-nio-8080-exec-1
inside asyncMethodTest: http-nio-8080-exec-1
```

@Aysn and Transaction Management

Usecase1: **Transaction Context do not transfer from caller thread to new thread which got created by Async.**

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```java
@RestController
@RequestMapping(value = "/api/")
public class UserController {

    @Autowired
    UserService userService;

    @PostMapping(path = "/updateuser")
    public String updateUserMethod(){
        userService.updateUser();
        return null;
    }
}
```

```java
@Component
public class UserService {

    @Autowired
    UserUtility userUtility;

    @Transactional
    public void updateUser(){

        //1. update user status
        //2. update user first name

        //3. update user
        userUtility.updateUserBalance();
    }
}
```

```java
@Component
public class UserUtility {

    @Async
    public void updateUserBalance(){
        //updating user balance amount.
    }
}
```

Usecase2: **Use with Precaution**, as new thread will be created and have transaction management too but context is not same as parent thread. So Propagation will not work as expected.

```java
@RestController
@RequestMapping(value = "/api/")
public class UserController {

    @Autowired
    UserService userService;

    @PostMapping(path = "/updateuser")
    public String updateUserMethod(){
        userService.updateUser();
        return null;
    }
}
```

```java
@Component
public class UserService {

    @Transactional
    @Async
    public void updateUser(){

        //1. update user status
        //2. update user first name
        //3. update user
    }
}
```

Usecase3: ✓

```java
@RestController
@RequestMapping(value = "/api/")
public class UserController {

    @Autowired
    UserService userService;

    @PostMapping(path = "/updateuser")
    public String updateUserMethod(){
        userService.updateUser();
        return null;
    }
}
```

```java
@Component
public class UserService {

    @Autowired
    UserUtility userUtility;

    @Async
    public void updateUser(){
        userUtility.updateUser();
    }
}
```

```java
@Component
public class UserUtility {

    @Transactional
    public void updateUser(){
        //1. update user status
        //2. update user first name
        //3. update user
    }
}
```

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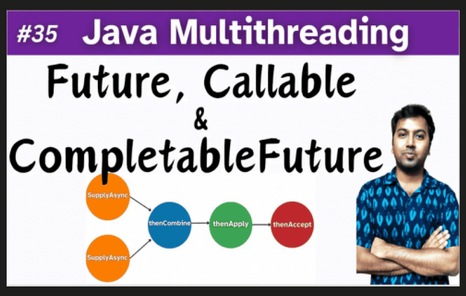
@Aysn Method return type

Both Future and Completable Future can be the return type of the Async method

Checkout Java Playlist to learn more in depth of Future and CompletableFuture



Using Future:

```java
@RestController
@RequestMapping(value = "/api/")
public class UserController {

    @Autowired
    UserService userService;

    @GetMapping(path = "/getuser")
    public String getUserMethod(){
       Future<String> result =  userService.performTaskAsync();
        String output = null;
       try {
           output = result.get();
           System.out.println(output);
       }catch (Exception e) {
            System.out.println("some exception");
       }
        return output;
    }
}
```

```java
@Component
public class UserService {

    @Async
    public Future<String> performTaskAsync(){
        return new AsyncResult<>( value: "async task result");
    }
}
```

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S.No.	Method Available in Future Interface	Purpose
1.	boolean cancel(boolean mayInterruptIfRunning)	• Attempts to cancel the execution of the task. • Returns false, if task can not be cancelled (typically bcoz task already completed); returns true otherwise.
2.	boolean isCancelled()	• Returns true, if task was cancelled before it get completed.
3.	boolean isDone()	• Returns true if this task completed. Completion may be due to normal termination, an exception, or cancellation -- in all of these cases, this method will return true.
4.	V get()	• Wait if required, for the completion of the task. • After task completed, retrieve the result if available.
5.	V get(long timeout, TimeUnit unit)	• Wait if required, for at most the given timeout period. • Throws 'TimeoutException' if timeout period finished and task is not yet completed.

Using CompletableFuture:

Introduced in Java8

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```java
@RestController
@RequestMapping(value = "/api/")
public class UserController {

    @Autowired
    UserService userService;

    @GetMapping(path = "/getuser")
    public String getUserMethod(){
       CompletableFuture<String> result =  userService.performTaskAsync();
        String output = null;
       try {
           output = result.get();
           System.out.println(output);
       }catch (Exception e) {
            System.out.println("some exception");
       }
        return output;
    }
}
```

```java
@Component
public class UserService {

    @Async
    public CompletableFuture<String> performTaskAsync(){
        return CompletableFuture.completedFuture( value: "async task result");
    }
}
```

Exception Handling

Method which has Return type

```java
@RestController
@RequestMapping(value = "/api/")
public class UserController {

    @Autowired
    UserService userService;

    @GetMapping(path = "/getuser")
    public String getUserMethod(){
        CompletableFuture<String> result =  userService.performTaskAsync();
         String output = null;
         try {
             output = result.get();
             System.out.println(output);
         }catch (Exception e) {
                System.out.println("some exception");
         }
          return output;
    }
}
```

During Get call, Exception is thrown and we can catch it and handled accordingly

```java
@Component
public class UserService {

    @Async
    public CompletableFuture<String> performTaskAsync(){
        return CompletableFuture.completedFuture( value: "async task result");
    }
}
```

Method which do not return anything

```java
@RestController
@RequestMapping(value = "/api/")
public class UserController {

    @Autowired
    UserService userService;

    @GetMapping(path = "/getuser")
    public String getUserMethod(){
        userService.performTaskAsync();
         return "";
    }
}
```

```java
@Component
public class UserService {

    @Async
    public void performTaskAsync(){
        //perform some task
    }
}
```

How to handle this?

1. Within Async Method itself

```java
@Component
public class UserService {

    @Async
    public void performTaskAsync(){
        try {
            //perform some task
        } catch (Exception e) {
            //hanlde the exception here
        }
    }
}
```

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2. Implement Custom AsyncExceptionHandler

```java
@Configuration
public class AppConfig implements AsyncConfigurer {

    @Autowired
    private AsyncUncaughtExceptionHandler asyncUncaughtExceptionHandler;

    @Override
    public AsyncUncaughtExceptionHandler getAsyncUncaughtExceptionHandler() {
        return this.asyncUncaughtExceptionHandler;
    }
}

@Component
class DefaultAsyncUncaughtExceptionHandler implements AsyncUncaughtExceptionHandler {

    @Override
    public void handleUncaughtException(Throwable ex, Method method, Object... params) {
        System.out.println("in default Uncaugh Exception method");
        //logging can be done here.
    }
}
```

```java
@Component
public class UserService {

    @Async
    public void performTaskAsync(){
        int i = 0;
        System.out.println(5/i);
    }
}
```

Output:

```
2024-08-18T20:15:14.082+05:30  INFO 70455 --- [nio-8080-exec-1]
in default Uncaugh Exception method
```

If , we will not handle it, then, Spring boot default SimpleAsyncUncaughtExceptionHandler will get invoked

```java
@Configuration
public class AppConfig  {

}
```

```java
@Component
public class UserService {

    @Async
    public void performTaskAsync(){
        int i = 0;
        System.out.println(5/i);
    }
}
```

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Spring boot framework code..

```java
public class SimpleAsyncUncaughtExceptionHandler implements AsyncUncaughtExceptionHandler {
    private static final Log logger = LogFactory.getLog(SimpleAsyncUncaughtExceptionHandler.class);

    public SimpleAsyncUncaughtExceptionHandler() {
    }

    public void handleUncaughtException(Throwable ex, Method method, Object... params) {
        if (logger.isErrorEnabled()) {
            logger.error( message: "Unexpected exception occurred invoking async method: " + method, ex);
        }

    }
}
```

Output:

```
 task-5] .a.i.SimpleAsyncUncaughtExceptionHandler : Unexpected exception occurred invoking async method:

java.lang.ArithmeticException Create breakpoint : / by zero
    at com.conceptandcoding.learningspringboot.AsyncAnnotationLearn.UserService.performTaskAsync(UserService.java:18)
```

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